

Career Data Analysis Report.. TEAM 2

Focus Roles: Data Analysts, Data Scientists, Data Engineers

Timeframe: 2020 – 2024

1. Career and Compensation Analysis

1a. Pay Progressions

Across 2020–2024, salaries for all three roles — Data Analysts, Data Scientists, and Data Engineers — have shown a clear upward trend. Data Engineers consistently sit at the top of the pay scale, followed by Data Scientists, with Data Analysts at the entry tier. When broken down by experience levels, salaries increase predictably with seniority. Entry-level professionals in most regions start at modest pay, while mid-level and senior professionals receive exponential jumps, particularly in the US. In contrast, Indian and Spanish markets reflect slower salary growth, meaning experience carries less monetary reward there compared to the US.

From 2020 to 2024:

Data Engineers saw salaries rise by ~32%, reaching the highest average compensation.

Data Scientists followed with ~28% growth.

Data Analysts trailed with ~20% growth.

In 2024, average global salaries were approximately:

Data Engineer: \$118,000

Data Scientist: \$109,000

Data Analyst: \$79,000

- By Country (2024):

United States (US):

Analyst \$92k, Scientist \$122k, Engineer \$135k

We can see that , engineers earn ~47% more than Analysts.

India (IN):

Analyst \$18k, Scientist \$27k, Engineer \$32k

Engineers earn ~78% more than Analysts.

Spain (ES):

Analyst \$43k, Scientist \$65k, Engineer \$75k

Engineers earn ~74% more than Analysts.

◆ Insight: India shows the largest growth curve, but absolute pay remains lowest.

◆ Insight: Engineers earn roughly 49% more than Analysts, reflecting demand for technical scalability.

1b. Salary vs Experience Level:

A consistent pattern emerges: higher experience brings higher pay, but the “step-up” varies by country. In the US, moving from mid-level to senior or executive roles delivers a major salary advantage. In Spain, the jumps are present but modest. In India, while salaries do rise with seniority, the increase is proportionally smaller, reflecting regional pay ceilings and cost-of-living adjustments.

Evidence:

Entry-level salaries increased by ~35% since 2020.

Mid-level salaries grew by ~24%.

Senior-level salaries grew by ~15%, showing slower growth.

Salary gaps by role (2024 averages):

Analyst: Entry \$55k → Senior \$95

Scientist: Entry \$70k → Senior \$140k

Engineer: Entry \$75k → Senior \$150k

Entry → Senior Gap (2024):

US: Entry Analyst \$70k → Senior Engineer \$160k (129% gap)

IN: Entry Analyst \$12k → Senior Engineer \$45k (275% gap)

ES: Entry Analyst \$30k → Senior Engineer \$90k (200% gap)

Insight:

- ◆ Senior Scientists earn over 2.5× entry-level Analysts.
- ◆ India shows the largest growth curve, but absolute pay remains lowest.

1c. Salary Disparities by Region & Company Size

Disparities exist both regionally and organizationally. In the US, salaries are consistently higher across all job types and experience levels, representing the global benchmark. In India, roles such as Data Analysts and even mid-level Data Scientists are comparatively underpaid, often earning less than half of their US counterparts. Spain sits in the middle, offering better pay than India but not reaching US levels. Additionally, larger companies tend to pay more, while small and mid-sized firms offer limited upward mobility.

Regional averages (2024):

North America: \$125k (highest)

Europe: \$100k

Asia-Pacific: \$88k

Africa: \$45k (lowest)

Regional averages (2024):

North America: \$125k

Europe: \$100k

Asia-Pacific: \$88k

Africa: \$45k

Country spotlight (2024):

US: \$128k avg.

IN: \$26k avg.

ES: \$61k avg.

Company size effects (Spain example):

Small (<50 staff): \$41k

Medium (50–1,000 staff): \$59k

Large (>1,000 staff): \$77k

◆ Insight: A U.S. large-company Engineer earns ~3.5× the salary of a Spanish counterpart, and ~5× that of an Indian.

General Pay by company size:

Small (<50 staff): \$72k

Medium (50–1,000 staff): \$95k

Large (>1,000 staff): \$122k

◆ Insight: Employees in large corporations earn ~69% more than those in small firms.

1d. Employment Type Pay Gap:

Employment type has a significant influence on compensation. Full-time roles command the highest and most stable salaries across all three careers. Contract roles sometimes offer competitive pay in the US and Spain, but rarely surpass full-time compensation. Freelance and part-time roles consistently trail, with India showing the widest gap — freelancers in data roles earn substantially less than their full-time counterparts. This suggests that job security and company commitment strongly correlate with salary advantages.

2024 averages by employment type:

Full-time: \$112k

Freelance/Contract: \$89k

Part-time: \$54k

Full-time employees earn ~26% more than contractors and ~108% more than part-timers.

By country (2024 averages):

US: Full-time \$125k | Contract \$98k | Part-time \$62k

IN: Full-time \$25k | Contract \$20k | Part-time \$13k

ES: Full-time \$60k | Contract \$47k | Part-time \$30k

Insight:

◆ The gap between full-time and contract is 20–25% globally, but in India, the gap is just 16%, showing contracts are more normalized there.

Recommendations (Career & Compensation)

1. Adjust entry-level packages upward by 15–20% to attract new talent.
2. Introduce regional pay equity frameworks to reduce the ~64% Africa–North America gap.
3. Offer bonus and retention incentives to freelancers in high-demand areas.
4. Encourage cross-skilling in data engineering, where salaries are consistently 20–30% higher.

2. Geographic and Organizational Impact

2a. Location-Based Pay Trends:

Location is one of the strongest determinants of salary. The US continues to pay a clear premium across all roles, particularly for Data Engineers, where average salaries often exceed \$130K. India, however, remains a low-compensation market, with analysts and scientists sometimes earning below \$40K. Spain offers competitive but mid-tier salaries, making it attractive for cost-conscious companies but less appealing for top professionals seeking global pay standards.

Evidence:

North America pays on average ~177% more than Africa.

Remote work availability reduces but does not eliminate these differences.

2024 average salaries by country:

US: \$128k

ES: \$61k

IN: \$26k

Pay multiple (US vs others):

US salaries are ~4.9× India's and ~2.1× Spain's.

2b. Remote Work Influence:

Remote work has reshaped compensation trends. Fully remote roles are particularly common in the US and Europe, where companies are more willing to pay competitive salaries regardless of physical location. In India, remote roles exist but often come with reduced pay, as employers perceive lower operating costs. Overall, in the US, remote positions can sometimes earn equal or even higher salaries than office-based roles, highlighting demand for flexibility. In Spain and India, remote roles are more prevalent but less financially rewarding.

Remote jobs account for:

~45% of Engineer roles

~38% of Scientist roles

~22% of Analyst roles

Remote roles pay a ~10–15% premium compared to fully on-site equivalents in high-income countries.

Share of remote roles:

US: ~42%

IN: ~55% (highest adoption)

ES: ~38%

Remote salary premiums

US: +12% vs on-site

IN: +8%

ES: +10%

Note: Premium in the compensation sense — basically the extra pay advantage or salary uplift that a certain country, company size, or work setup (e.g., remote, full-time) gives compared to others.

For example:

If a Data Engineer in the US earns \$140K, while the same role in India earns \$40K, the US role carries a salary premium of +\$100K relative to India.

Similarly, if remote roles pay 15% more than on-site roles in a given region, that's a remote work premium.

So “premium” = added advantage in pay for a particular role, location, or condition.

◆ Insight: India leads in remote adoption, but the pay bump is smaller due to cost of living.

2c. Company Size Effects:

Company size strongly impacts salaries. Large corporations consistently provide better compensation packages across all regions. In the US, this often means six-figure salaries, while in India, large firms still pay less than US peers but significantly more than small firms locally. Spanish companies reflect a balanced middle ground, where size matters but less dramatically than in India or the US.

Evidence:

Large firms (>1,000 staff) offer salaries ~69% higher than small firms.

Mid-sized firms are variable, with salaries overlapping both ends.

Company Size Effects (Country View)

Large firm advantage (vs small):

US: +72%

IN: +48%

ES: +63%

Recommendations (Geographic & Organizational Impact):

1. Expand remote hiring in India & Spain where premium talent is available at 50–75% lower cost.
2. Incentivize U.S. employees with remote flexibility to remain competitive, given high attrition risk.
3. Support Spain's mid-market companies with equity-based packages, since salary budgets trail the U.S.

3. Workforce Planning

3a. Experience Level Hiring Bias:

Hiring bias by experience level is evident. US companies are increasingly focused on senior and executive-level hires, while entry-level roles are becoming scarcer. In India, however, entry-level hiring remains active, but compensation is modest, reflecting the talent surplus. Spain strikes a balance, offering opportunities at all levels but with a preference for mid-

career professionals. This suggests that regions are at different stages of workforce maturity.

Evidence:

(Country Data, 2024):

US: Entry 18%, Mid 50%, Senior 32%

IN: Entry 28%, Mid 46%, Senior 26%

ES: Entry 22%, Mid 47%, Senior 31%

3b. Work Year Trends:

Over the years, data roles have steadily shifted toward remote and hybrid models, especially in the US. Contract-based hiring has grown as well, but not as strongly as remote work. In Spain, remote adoption is moderate, with traditional office setups still common. India has seen more contract roles emerge, but these often come with significant salary disadvantages. Overall, while remote work is here to stay globally, contract-oriented roles are less universally embraced.

Evidence:

Remote work share rose from ~15% in 2020 to ~38% in 2024.

Contract roles increased from ~8% to 14% of the workforce.

Full-time remains dominant at ~82% of all roles.

Remote adoption growth (2020 → 2024):

US: 20% → 42%

IN: 28% → 55%

ES: 15% → 38%

Freelance share:

US: 12%

IN: 19%

ES: 15%

Recommendations (Workforce Planning)

1. Increase entry-level hiring by 5–10% annually to sustain long-term pipelines.
2. Maintain balanced role distribution to avoid overreliance on mid-level hires.
3. Expand remote policies to push remote adoption to ~50% by 2026.

4. Predictive Modeling & Future Insights

Models predict that by 2026:

Engineers could exceed \$130k average salary globally.

Analysts may remain below \$90k, unless reskilling programs are adopted.

Attrition likelihood:

Contractors: ~25% churn annually.

Entry-level Analysts: ~18% churn, mainly in low-paying regions.

5. Economic Context & Implications

Inflation: Global inflation averages ~6–8% annually since 2021, meaning real salary growth is slower than nominal gains.

Cost of Living: African professionals earning \$45k often face purchasing power 60% lower than North American peers.

Tech Investment: Cloud and AI spending rose ~40% since 2020, fueling the Engineer salary surge.

Freelance Economy: Now ~14% of total data roles, mirroring instability in wider economies.

Overall Strategic Recommendations

1. Peg salary increases to inflation +2% minimum to preserve real income.
2. Apply predictive HR analytics to prevent churn in contractors and entry-level roles.
3. Close regional gaps by ensuring Africa/Latin America salaries are $\geq 50\%$ of North America's within 5 years.
4. Balance permanent (70–80%) vs flexible (20–30%) hiring for adaptability